Inverted MAML Card 1: Round 19 Thermal Bit-Slicing Report

Hardware: Card 1 (T120+T20, 1k/2.2k) | **FW:** v2.2-MAML (0x20261030) | **Params:** 40% PWM, K=1, LR=0.055, L2=1920, Mom=0.85, Clamp=±90 LSB

1. Round 19 Summary & Cumulative Comparison (R1–R18)

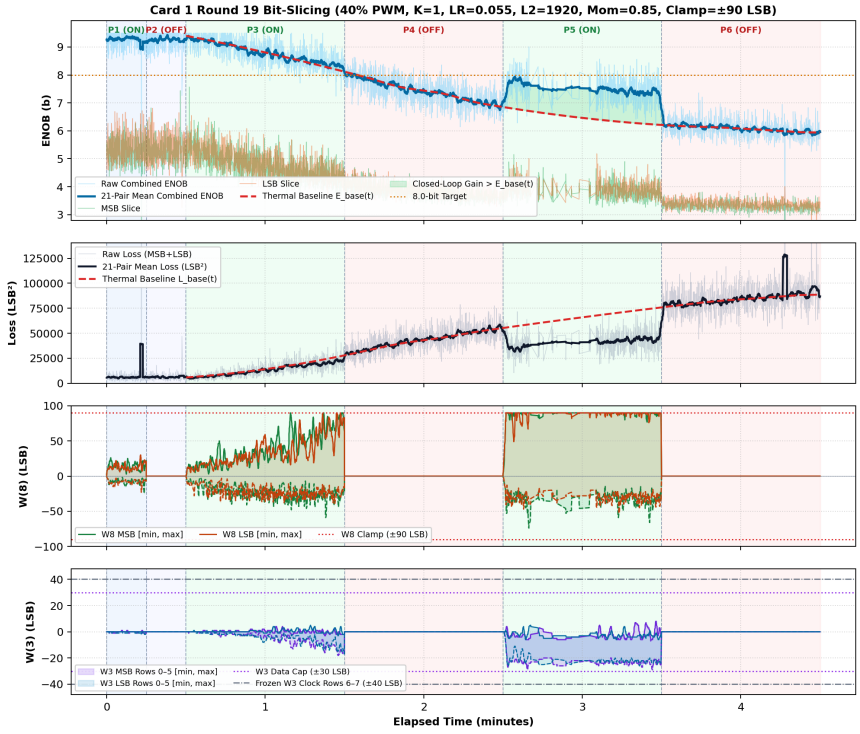
• **Cold Verification:** Phase 1 ON = **9.45 b**, Phase 2 OFF = **9.45 b** (W8=0 strictly).

• **Round 19 Detrended Trend Metrics:** Lift vs E_base = **+0.77 b**, Loss Ratio = **1.52x**, Climb = **+0.74 b**, Jitter = **0.37 b**, Recovery = **0.67**, Score = **+1.888**.

Rnd	LR	L2	Mom	Clamp	Lift	LossR	Climb	Jitt	Recov	Score
R11dc_fix	0.055	1920	0.85	±80	+0.95b	1.70x	+0.98b	0.42b	0.00	+1.810
R12dc_fix	0.055	1920	0.85	±82	+1.04b	1.75x	+0.83b	0.42b	0.00	+1.721
R13dc_fix	0.055	1920	0.85	±85	+1.12b	1.78x	+1.06b	0.43b	0.50	+2.645
R14dc_fix	0.055	1920	0.85	±90	+0.78b	1.55x	+0.97b	0.39b	0.67	+2.117
R15dc_fix	0.055	1920	0.85	±100	+0.94b	1.67x	+0.89b	0.40b	0.75	+2.443
R16dc_fix	0.055	1920	0.85	±90	+0.70b	1.52x	+0.82b	0.34b	0.75	+2.026
R17dc_fix	0.055	1920	0.85	±90	+0.88b	1.65x	+1.25b	0.39b	0.50	+2.385
R18dc_fix	0.055	1920	0.85	±90	+0.77b	1.52x	+0.74b	0.37b	0.67	+1.888

2. Phase-by-Phase Telemetry (3,924 Pairs)

Phase	Regime	N	Start->Late (Mean)	Δ vs E_base	Loss Start->Late	W8 Pk [M,L]	W3 Pk	W8 nz
P1	Cold / Regler ON	322	9.24->9.21 (9.26b)	-	6408->12249	[26,30]	2	322
P2	Cold / Regler OFF	294	9.20->9.26 (9.27b)	-	5932->6428	[0,0]	0	0
P3	Warm / Regler ON	1046	9.35->8.46 (8.80b)	+0.08 b	5618->19307	[90,90]	18	1046
P4	Warm / Regler OFF	920	7.96->7.07 (7.44b)	+0.01 b	30475->51289	[0,0]	0	0
P5	Warm / Regler ON	559	7.15->7.37 (7.49b)	+1.11 b	49449->44113	[90,90]	29	559
P6	Warm / Regler OFF	783	6.23->5.97 (6.08b)	-0.00 b	78219->91740	[0,0]	0	0



3. All W(8) (7x6 MSB+LSB) & W(3) (8x6 MSB+LSB) Weight Dynamics — Round 19

• **Full 180-Synapse Per-Pair Capture:** Every individual weight in **Matrix 8 MSB (42 weights)**, **Matrix 8 LSB (42 weights)**, **Matrix 3 MSB (36 active + 12 clock)**, and **Matrix 3 LSB (36 active + 12 clock)** is plotted below across all columns `Col 0..5`. W(8) enforces anti-windup clamping at ±90 LSB, while W(3) rows 0–5 stay within ±30 LSB so rows 6–7 clock staircases remain locked at ±40 LSB.

Round 19 Full Hardware Weight Telemetry: ALL 84 W(8) [7x6 MSB+LSB] & ALL 96 W(3) [8x6 MSB+LSB] Synapses

